

# D - Not Adjacent 2

分值: 400 分

## Problem Statement

You are given a string  $S$  consisting of a, b, c.

给定一个仅由 a、b、c 组成的字符串  $S$ 。

Find the number of non-empty **subsequences** of  $S$  in which no two adjacent characters are the same, modulo 998244353.

请计算  $S$  的满足「不存在两个相邻字符相同」的非空**子序列**的数量，结果对 998244353 取模。

Two subsequences are considered distinct if they are taken from different positions, even if they are identical as strings.

只要选取的位置不同，即使对应的字符串内容完全相同，也视为不同的子序列。

**What is a subsequence?**

**什么是子序列？**

A **subsequence** of  $S$  is a string obtained by removing zero or more characters from  $S$  and concatenating the remaining characters in their original order. For example, ab, ac are subsequences of abc, but ca, bb are not subsequences of abc.

字符串  $S$  的**子序列**是指：从  $S$  中删除零个或多个字符后，将剩余字符按原顺序拼接得到的字符串。例如 ab、ac 是 abc 的子序列，但 ca、bb 不是 abc 的子序列。

## Constraints

- $S$  is a string of length between 1 and  $3 \times 10^5$ , inclusive, consisting of a, b, c.
- $S$  是长度在 1 到  $3 \times 10^5$  之间（含两端）的字符串，仅由 a、b、c 组成。

## Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

$S$

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## Output

Output the answer.

输出答案。

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## Sample Input 1

abbc

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## Sample Output 1

11

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The subsequences in which no two adjacent characters are the same are the following 11:

满足相邻字符互不相同的子序列共有以下 11 个：

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- a (the 1st character of  $S$ )
- b (the 2nd character of  $S$ )
- b (the 3rd character of  $S$ )
- c (the 4th character of  $S$ )
- ab (the 1st, 2nd characters of  $S$ )
- ab (the 1st, 3rd characters of  $S$ )
- ac (the 1st, 4th characters of  $S$ )
- bc (the 2nd, 4th characters of  $S$ )
- bc (the 3rd, 4th characters of  $S$ )

- abc (the 1st, 2nd, 4th characters of  $S$ )
- abc (the 1st, 3rd, 4th characters of  $S$ )
- a (取自  $S$  的第 1 个字符)
- b (取自  $S$  的第 2 个字符)
- b (取自  $S$  的第 3 个字符)
- c (取自  $S$  的第 4 个字符)
- ab (取自  $S$  的第 1、2 个字符)
- ab (取自  $S$  的第 1、3 个字符)
- ac (取自  $S$  的第 1、4 个字符)
- bc (取自  $S$  的第 2、4 个字符)
- bc (取自  $S$  的第 3、4 个字符)
- abc (取自  $S$  的第 1、2、4 个字符)
- abc (取自  $S$  的第 1、3、4 个字符)

Note that, as with the 2nd and 3rd entries, two subsequences are considered distinct if they are taken from different positions, even if they are identical as strings.

注意，正如第 2 项和第 3 项所示，只要选取的位置不同，即使字符串内容相同，也视为不同的子序列。

## Sample Input 2

```
cabcabcbccacbcbaabacbacabccacbccbcacbacbacabacacabccacccaaaaabababcbabacaccabbccacbc
```

## Sample Output 2

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378217423
```

Output the count modulo 998244353.

输出结果对 998244353 取模。